

Outside Good Uncertainty

Ordinal Dual Response in Choice-Based Conjoint Analysis

Dan Yavorsky Geoffery Zheng

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We present a behavioral model in which consumers identify their most-preferred option among a set of alternatives but face uncertainty in their evaluation of the outside good. The model rationalizes the dual-response format used in choice-based conjoint analysis and extends the follow-up purchase question from binary to ordinal. The resulting likelihood function combines a multinomial logit for the forced choice with a cumulative-link ordinal model driven by the task’s inclusive value. The link is logistic or Gumbel, depending on whether the respondent resolves her uncertainty before answering, and the binary dual-response model is a special case. We derive how much information a finer response scale adds, provide estimation routines for both aggregate and hierarchical Bayesian versions of the model, and show that the model outperforms both dichotomizing the scale into a binary response and assigning fixed purchase probabilities to each scale point. We apply the model to a choice-based conjoint study conducted in partnership with a consumer insights consultancy.

1 Introduction

Consumers are good at indicating which product they like best. They are often less certain about whether they would buy it. Choice-based conjoint analysis, the workhorse of stated-preference measurement in marketing (Allenby et al. 2019), has a device for separating these two: the dual-response format (Brazell et al. 2006). In each task, the respondent first makes a forced choice among the displayed product profiles and then answers a follow-up question about whether she would actually purchase her selection. This format was introduced to address a practical problem with the traditional no-choice option: when “none of these” appears alongside the profiles in a traditional conjoint task, each time it is chosen contributes nothing to estimating attribute preferences. The dual-response design recovers that lost information because the forced choice always reveals preference among profiles, while still measuring demand for the category through the follow-up question, without systematically biasing the preference estimates (Brazell et al. 2006; Dhar and Simonson 2003). Following its implementation in

Sawtooth Software’s widely used choice-based conjoint (CBC) system (Diener et al. 2006; Sawtooth Software 2017), the dual-response format became a common practice in commercial conjoint research. However, the dual-response format was not derived from a behavioral model of the respondent, and any such model has to take a stand on what she is uncertain about when the follow-up question arrives. This paper supplies such a model for an ordinal follow-up. It rationalizes the dual-response format, determines the link function rather than assuming one, and extends the purchase question from binary to ordinal.

We start from the behavioral premise that a respondent facing a dual-response conjoint task can indicate her most-preferred product profile in the forced choice, but whether she would purchase her selection is a question pertaining to a hypothetical future occasion about which she remains genuinely uncertain. That second question has been canonically binary: “would you buy the option you selected above?”, answered yes or no and modeled by pairing a multinomial logit for the forced choice with a binary logit for the purchase question (Diener et al. 2006; Hung, Liu, et al. 2025). But a yes-or-no answer requires the respondent to settle something she has not yet had reason to settle. What she holds is a probability (Manski 1990). A finer scale would ask instead for her degree of conviction, and carry more information about what she will do. The natural extension is therefore an *ordinal* dual-response follow-up, “how likely are you to buy the option you selected above?”, answered, e.g., on a five-, seven-, or eleven-point scale from “very unlikely” to “very likely.”

We formalize that uncertainty as a component of utility for the outside good that she has not yet resolved, and, in our primary specification, we let the error components (for both the inside goods and the outside good) be Gumbel (McFadden 1981; Train 2009). (Appendix D relaxes independence across goods to any generalized extreme value structure.) Together the behavioral premise and distributional kernel do two things for the model. First, they deliver a closed-form likelihood: the maximum of Gumbel utilities is itself Gumbel, with location equal to the log-sum inclusive value, and independent of which alternative attains it, so the two responses factor and the model pairs a multinomial logit for the forced choice with a cumulative link for the ordinal report. Second, they determine the link to be complementary log-log. The familiar ordered logit (logistic link) arises instead when the uncertainty is resolved before the follow-up, which is the assumption underlying the binary dual-response models and is stated explicitly by Hung, Liu, et al. (2025). At two scale points the logistic link is the “Unified Model” of Hung, Liu, et al. (2025), equivalently the DR-AnyMax model of Diener et al. (2006) and, once the inclusive value cancels, the single-stage multinomial logit with a no-choice constant (Haaijer et al. 2001). Our model predicts a complementary log-log binary response, so the familiar yes/no follow-up is mildly misspecified whenever respondents remain uncertain.

We make four contributions. First, we develop the behavioral model: a utility-theoretic account of dual-response conjoint in which outside-good uncertainty is the primitive. That premise rationalizes the design and fixes the link function, and the cumulative-link family it generates contains the existing binary models as the two-point case of the alternative link, so the assumption those models rest on becomes explicit and testable. Second, we analyze the information content of the ordinal follow-up: we derive the information a second response of

a given length contributes and show that it is always non-negative and never falls when the scale is made finer, extending the binary efficiency results of Brazell et al. (2006) and Hung, Liu, et al. (2025) and providing design guidance on how many scale points to use. Third, we provide estimation routines: maximum likelihood for the aggregate model and a custom Markov chain Monte Carlo sampler for the hierarchical Bayesian model with heterogeneous preferences, including an extension with respondent-specific cut points that accommodates individual differences in scale usage (Greene and Hensher 2010; Falco et al. 2015). We also express the model in Apollo (Hess and Palma 2019) both as a check on our implementation and as a route for readers already working in that framework. Fourth, we supply evidence, including simulation studies that demonstrate parameter recovery and quantify the losses from dichotomization and constant-probability weighting, as well as an empirical application to a commercial choice-based conjoint study with an ordinal dual response, conducted in partnership with a consumer insights consultancy. Complete replication code accompanies the paper, with each study in the simulation program laid out as an executable notebook.

Several literatures stand behind the model. Marketing work on dual-response and no-choice designs established why asking two questions beats adding “none of these” to the list of products: the preference information in a task survives even when the respondent would not buy. That work set the conventions for estimating such models (Brazell et al. 2006; Diener et al. 2006; Haaijer et al. 2001; Hung, Liu, et al. 2025), and it continues to refine the design. Schlereth and Skiera (2017) separate the forced and the free questions into blocks and add an adaptive rule for choosing which free questions to ask, which limits context effects and extreme response behavior when willingness to pay is the target. The appeal of the no-buy option also shifts as a respondent works through a long survey (Hung, Kurz, et al. 2025). The opt-out literature in choice modeling documents the consequences of getting the no-buy option wrong: predicted market shares and estimates of the value consumers get from a product both come out distorted (Campbell and Erdem 2019).

The separation of the two decisions is older than the dual-response design. Demand models fit to supermarket scanner data have long estimated whether a household buys in a category separately from which brand it picks, joining the two stages through the inclusive value of the brand-choice stage (Gupta 1988; Bucklin and Gupta 1992). Our second response uses that device on survey data. Our statistical tools come from work on ordered and mixed response formats, from frameworks that handle ratings, rankings, and choices together (Marshall and Bradlow 2002) to ordered-choice models that let different people use the same scale differently (Greene and Hensher 2010; Falco et al. 2015). A further literature explains why the follow-up should ask for a probability. A yes/no question forces a rounding to zero or one, and Manski (2004) argues that economists should elicit the probability itself. Marketing research made the same case earlier: Juster (1966) replaced the yes/no buying-intentions question with an eleven-point probability scale, and Morrison (1979) showed that stated intentions predict purchases only probabilistically, and that the mapping from one to the other depends on context. To our knowledge, no prior work derives the forced choice and the ordinal purchase answer from a single account of the consumer.

The remainder of the paper proceeds as follows. Section 2 develops the behavioral model, derives the joint likelihood, establishes the nesting relationships, and presents the information-content and estimation results. Section 3 reports the simulation study. Section 4 presents the empirical application. Section 5 discusses implications for practice, including guidance on scale design, and Section 6 concludes. Appendix A collects the proofs, Appendices B and C give the sampler and marginal-likelihood details, Appendix D extends the model to choice sets whose alternatives are correlated, and Appendix E gives the implementation in Apollo.

2 Model

2.1 Setup

Consumer $i = 1, \dots, N$ faces choice tasks $t = 1, \dots, T$. Each task presents a set of “inside” goods $\mathcal{J}^+ = \{1, 2, \dots, J\}$ alongside an “outside” good $j = 0$, so that the full choice set is $\mathcal{J} = \{0, 1, 2, \dots, J\}$.¹ Consumer i derives utility u_{itj} from good j in task t :

$$u_{itj} = h(\mathbf{x}_{tj}, \beta_i) + \eta_{itj}, \quad (1)$$

where $\mathbf{x}_{tj}$ is a length- P column vector of good characteristics, β_i is a length- P column vector of consumer-specific taste parameters, and η_{itj} encapsulates factors known to the consumer but unobserved by the researcher. We take $h(\mathbf{x}_{tj}, \beta_i) = \mathbf{x}_{tj}'\beta_i$ and write $V_{itj} := \mathbf{x}_{tj}'\beta_i$ for this deterministic component, but the linear specification is not required.

The outside good is special in two respects. First, it is associated with a zero vector of characteristics ($\mathbf{x}_{t0} = \mathbf{0}$), so that $V_{it0} = 0$; this is the standard normalization that anchors the location of utility (Haaijer et al. 2001). Second, and this is the paper’s central behavioral assumption, the consumer observes η_{itj} for all inside goods but does not observe η_{it0} . She is comparing goods that are in front of her now; the outside good stands in for a purchase occasion that has not yet arrived, and her evaluation of it is uncertain.^{2 3}

In each task the consumer provides two responses:

1. **First response (choice).** She reports her preferred inside good,

$$j_{it}^* = \operatorname{argmax}_{j \in \mathcal{J}^+} u_{itj}. \quad (2)$$

¹We hold J fixed across tasks and consumers to lighten notation; nothing in what follows requires it.

²This can be motivated by a framework in which $u_{it0} = 0$ and $u_{itj} = \mathbf{x}_{tj}'\beta_i + \eta_{itj} - \eta_{it0}$ for $j \in \mathcal{J}^+$, where η_{it0} captures the consumer’s uncertainty about her future tastes or circumstances at the moment of purchase. Because utilities are ordinal and η_{it0} is a common shock, it plays no role in the comparison among inside goods.

³While the parametrization $\eta_{it0} \sim \text{Gumbel}(0, 1)$ preserves symmetry among the $J + 1$ goods and is thus a natural choice, the framework easily accommodates alternative distributions for η_{it0} . For example, an affine function of individual characteristics could accommodate individual-level variation in the propensity to prefer the outside good.

This is a forced choice: the outside good is not among the options.

2. **Second response (ordinal purchase evaluation).** She reports a value y_{it} on a discrete ordinal scale $w \in \mathcal{W} = \{1, \dots, W\}$ (e.g., “very unlikely” to “very likely” on a five-point scale) that characterizes her evaluation of purchasing j_{it}^* , that is, of choosing it over the outside good. The binary dual-response format of Brazell et al. (2006) and Diener et al. (2006) is the special case $W = 2$ of this design.

We model $\eta_{itj} \stackrel{iid}{\sim} \text{Gumbel}(0, 1)$, independent of $\mathbf{x}_{tj}$ and independent across goods, tasks, and consumers. This is the logit kernel standard in discrete choice (McFadden 1981; Train 2009), inherited by choice-based conjoint. The independence across goods is stronger than our derivation needs. Appendix D shows that Lemma 2.1 holds for any generalized extreme value error structure once S_{it} is replaced by the GEV generator, so the following results carry over to correlated alternatives, but for exposition we proceed with the independent errors assumption.

Two classical properties of the Gumbel family are used throughout (McFadden 1981; Cardell 1997).

Lemma 2.1. *Let $u_{itj} = V_{itj} + \eta_{itj}$ with $\eta_{itj} \stackrel{iid}{\sim} \text{Gumbel}(0, 1)$ for $j \in \mathcal{J}^+$, and define $u_{it}^* = \max_{j \in \mathcal{J}^+} u_{itj}$. Then*

(i) $u_{it}^* \sim \text{Gumbel}(\bar{\mu}_{it}, 1)$ with location

$$\bar{\mu}_{it} = \ln S_{it}, \quad S_{it} := \sum_{j \in \mathcal{J}^+} \exp(V_{itj}); \quad (3)$$

(ii) *the identity of the maximizer j_{it}^* and the value of the maximum u_{it}^* are statistically independent.*

Property (ii) makes the dual-response likelihood tractable: whatever the consumer tells us about the level of her best utility (the second response) is independent of which good delivered it (the first response), conditional on the deterministic utilities. Proofs are in Appendix A.

2.2 First Response: Multinomial Logit

The first response is a standard forced choice among the inside goods. Under the Gumbel assumption,

$$\Pr(j_{it}^* = j) = \frac{\exp(V_{itj})}{\sum_{k \in \mathcal{J}^+} \exp(V_{itk})} = \frac{\exp(V_{itj})}{S_{it}}, \quad (4)$$

the multinomial logit over inside goods only. Note that the outside good does not appear: the forced choice does not depend on the consumer’s uncertain evaluation of the outside good, which is the design’s appeal (Brazell et al. 2006).

2.3 Second Response: An Ordinal Model of the Purchase Evaluation

What does the consumer consult when answering “how likely are you to purchase the option you selected?” Our premise answers the question: she has not resolved η_{it0} , so what she has to report is a probability. That premise delivers a likelihood of *cumulative-link* form, depending on the choice set only through the inclusive value $\bar{\mu}_{it}$, and it determines the link. We derive that model first. We then ask what follows if she instead resolves the shock before answering, as Hung, Liu, et al. (2025) assume. For a binary follow-up that assumption gives their Unified Model. Carried into an ordinal follow-up it is harder to interpret, for reasons we take up below.

2.3.1 Reporting a Purchase Probability

The consumer carries her uncertainty about the outside good into her answer and reports a probability. She knows u_{it}^* (she just identified j_{it}^*) but does not know η_{it0} , so from her perspective the probability that she would follow through and purchase j_{it}^* is

$$p_{it} = \Pr(\eta_{it0} < u_{it}^* \mid u_{it}^*) = F(u_{it}^*), \quad (5)$$

where $F(z) = \exp(-e^{-z})$ is the standard Gumbel CDF. The ordinal scale discretizes the unit interval: fix cut points $0 = \alpha_0 < \alpha_1 < \dots < \alpha_{W-1} < \alpha_W = 1$ and suppose the consumer reports

$$y_{it} = w \iff p_{it} \in [\alpha_{w-1}, \alpha_w). \quad (6)$$

We assume consumers share the definitions of the qualitative scale labels, so the α_w are common across consumers (we relax this in Section 2.10.1); the interior cut points are unobserved by the researcher and are estimated.

Reporting $y_{it} = w$ is equivalent to reporting that $u_{it}^* \in [F^{-1}(\alpha_{w-1}), F^{-1}(\alpha_w))$, an interval statement about the consumer’s best utility. From the researcher’s perspective, $u_{it}^* \sim \text{Gumbel}(\bar{\mu}_{it}, 1)$ by Lemma 2.1(i), so

$$\begin{aligned} \Pr(y_{it} = w) &= F_{\text{Gumbel}(\bar{\mu}_{it}, 1)}(F^{-1}(\alpha_w)) - F_{\text{Gumbel}(\bar{\mu}_{it}, 1)}(F^{-1}(\alpha_{w-1})) \\ &= \alpha_w^{S_{it}} - \alpha_{w-1}^{S_{it}}, \end{aligned} \quad (7)$$

using $F_{\text{Gumbel}(\bar{\mu}, 1)}(c) = \exp(-e^{-(c-\bar{\mu})}) = \exp(e^{\bar{\mu}} \ln \alpha) = \alpha^S$ when $c = F^{-1}(\alpha) = -\ln(-\ln \alpha)$. The second-response probabilities are differences of powers of the cut points, with the exponent $S_{it} = e^{\bar{\mu}_{it}}$ summarizing the attractiveness of the task’s inside goods: the more attractive the set, the more probability mass shifts toward high scale points.⁴

⁴Equivalently, from the researcher’s perspective the consumer’s purchase probability is Beta distributed, $p_{it} \sim \text{Beta}(S_{it}, 1)$. The Gumbel family is max-stable, so a shift in location is a power of the distribution function, $F_{\text{Gumbel}(\bar{\mu}, 1)}(z) = F(z)^S$ with $S = e^{\bar{\mu}}$, and therefore $\Pr(p_{it} \leq a) = \Pr(u_{it}^* \leq F^{-1}(a)) = a^{S_{it}}$, which is the $\text{Beta}(S, 1)$ distribution function on $[0, 1]$. Equation 7 is that distribution function differenced across the cut points. Its mean, $E[p_{it}] = S_{it} / (S_{it} + 1) = \Lambda(\bar{\mu}_{it})$, is the factor that appears in the conditional expectation of Section 2.5.

2.3.2 The Resolved Comparison, and the Binary Incumbent

Suppose instead that the consumer resolves the outside-good shock before answering and evaluates the realized utility difference

$$d_{it} = u_{it}^* - \eta_{it0}.$$

The binary question asks only for the sign of d_{it} . Here she grades its magnitude on the W -point scale: fix utility-scale cut points $-\infty = c_0 < c_1 < \dots < c_{W-1} < c_W = \infty$ and suppose

$$y_{it} = w \iff d_{it} \in [c_{w-1}, c_w). \quad (8)$$

From the researcher's perspective, $u_{it}^* \sim \text{Gumbel}(\bar{\mu}_{it}, 1)$ and $\eta_{it0} \sim \text{Gumbel}(0, 1)$ are independent, so their difference is logistic: $d_{it} \sim \text{Logistic}(\bar{\mu}_{it}, 1)$. Hence

$$\Pr(y_{it} = w) = \Lambda(c_w - \bar{\mu}_{it}) - \Lambda(c_{w-1} - \bar{\mu}_{it}), \quad \Lambda(z) = \frac{1}{1 + e^{-z}}. \quad (9)$$

This is an ordered logit (Greene and Hensher 2010) with the inclusive value $\bar{\mu}_{it}$ as the (negatively signed) linear predictor. The assumption that she resolves the shock before answering is what underlies the binary dual-response models. Hung, Liu, et al. (2025) state it explicitly, specifying that respondents bring the same private information to both responses, and derive the binary case from it; Diener et al. (2006) reach the same functional form by a different route. It is not our model, but it is the specification the existing binary literature uses, and Section 2.6 shows why that use is well founded at $W = 2$.

Both accounts therefore share a form. Define $c_w = F^{-1}(\alpha_w)$ so that the probability-scale cut points of Equation 6 sit on the utility scale alongside those of Equation 8. The second response is then a cumulative link in the inclusive value,

$$\Pr(y_{it} \leq w) = G(c_w - \bar{\mu}_{it}), \quad G = F \text{ (standard Gumbel)}, \quad (10)$$

the complementary log-log link. Replacing G with the logistic Λ gives Equation 9, which we invoke only at $W = 2$, where it is the incumbent binary model (Section 2.6). We write the derivations in terms of a general G because they do not depend on its form.

2.4 Joint Likelihood

By Lemma 2.1(ii), the value of u_{it}^* is independent of the identity of j_{it}^* ; and η_{it0} is independent of the inside-good components. The two responses in a task are therefore independent conditional on $(\mathbf{x}_t, \beta_i)$, and the per-task likelihood contribution is simply the product of Equation 4 and Equation 10:

$$\ell_{it}(\beta_i, \mathbf{c}) = \underbrace{\frac{\exp(V_{itj^*})}{S_{it}}}_{\text{choice (MNL)}} \times \underbrace{\left[G(c_{y_{it}} - \bar{\mu}_{it}) - G(c_{y_{it}-1} - \bar{\mu}_{it}) \right]}_{\text{ordinal purchase evaluation}}, \quad (11)$$

with $\mathbf{c} = (c_1, \dots, c_{W-1})$ and the full likelihood $L_i = \prod_t \ell_{it}$. This factorization is a consequence of Lemma 2.1(ii), the independence of the argmax and the maximum.

2.5 The Model in Miniature

Before turning to special cases, we make the machinery concrete with the smallest interesting instance of the model: two respondents with identical preferences, one attribute, one task each, and a three-point scale. The example is deliberately tiny, but it exhibits the model’s central mechanism.

Let both respondents share the scalar taste parameter $\beta = 1$, which we treat as known for the arithmetic. Each faces one task with two inside goods described by a single attribute x , answers the forced choice, and then reports on the three-point scale $\mathcal{W} = \{\text{"unlikely"}, \text{"unsure"}, \text{"likely"}\}$ with utility-scale cut points $\mathbf{c} = (c_1, c_2) = (0, 2)$. (In our model these correspond to probability-scale cut points $(\alpha_1, \alpha_2) = (F(0), F(2)) = (0.37, 0.87)$: the consumer answers “unsure” when her purchase probability lies between 37% and 87%.) The two tasks (one facing each respondent) differ only in the level of the attribute and not in its spread: consumer 1 faces an attractive pair, $x = (2, 1)$, and consumer 2 an unattractive pair, $x = (-1, -2)$.

Consider the model’s account of their behavior, giving both consumers the *same* realized components $\eta = (0.30, -0.86)$ on their two inside goods. Consumer 1’s utilities are $u = (2.30, 0.14)$: she picks her first good and knows her best utility, $u^* = 2.30$. She cannot resolve the outside-good shock η_0 , so she reports the bin containing her purchase probability $p = F(2.30) = 0.90$, namely “likely.” Consumer 2’s utilities are $u = (-0.70, -2.86)$: she also picks her first good, but her best utility is $u^* = -0.70$, her purchase probability is $F(-0.70) = 0.13$, and she reports “unlikely.” The two consumers made identical first-response choices from identical realizations, yet gave opposite second responses.

The walkthrough above fixed one set of draws. Averaging over all of them, as the researcher must, we see that the pattern is not an accident of that particular draw (see Table 1). The forced-choice probabilities are the same function of the data for both consumers: the MNL depends only on the within-task difference $x_1 - x_2 = 1$, so $\Pr(\text{choose better good}) = \Lambda(\beta) = 0.73$ in both tasks, and no quantity of forced-choice data could ever distinguish them. The tasks separate only through the inclusive value ($S_1 = e^2 + e^1 = 10.11$ versus $S_2 = e^{-1} + e^{-2} = 0.50$, i.e., $\bar{\mu}_1 = 2.31$ versus $\bar{\mu}_2 = -0.69$), which the cumulative-link stage compares against the cut points. (With $c_1 = 0$, consumer 2’s probability of “unlikely” is $\Lambda(-\bar{\mu}_2) = 1/(1 + S_2) = 0.67$, the familiar no-choice probability of the single-stage MNL, which previews the nesting results below.)

Table 1: Two tasks that are indistinguishable in the forced choice and nearly opposite in the ordinal report ($\beta = 1$, $\mathbf{c} = (0, 2)$, $W = 3$).

	Consumer 1 (attractive task)	Consumer 2 (unattractive task)
Attribute values x	(2, 1)	(−1, −2)
$\Pr(\text{choose better good})$	0.73	0.73
$S = \sum_j e^{V_j}$,	10.11, 2.31	0.50, −0.69
$\bar{\mu} = \ln S$		

	Consumer 1 (attractive task)	Consumer 2 (unattractive task)
$\Pr(y = w)$	(0.00, 0.25, 0.75)	(0.60, 0.33, 0.07)
$E[p \mid y = \text{"unsure"}]$	0.80	0.60

The joint likelihood of the observed data, in which each consumer chose her better good and $y_1 = \text{"likely"}$, $y_2 = \text{"unlikely"}$, follows from Equation 11, and is the same expression under either link:

$$L(\beta, c_1, c_2) = \underbrace{\Lambda(\beta) [1 - G(c_2 - \bar{\mu}_1(\beta))]}_{\text{consumer 1: } 0.73 \times 0.75} \times \underbrace{\Lambda(\beta) G(c_1 - \bar{\mu}_2(\beta))}_{\text{consumer 2: } 0.73 \times 0.60}, \quad (12)$$

with $\bar{\mu}_1(\beta) = \beta + \ln(1 + e^\beta)$ and $\bar{\mu}_2(\beta) = -2\beta + \ln(1 + e^\beta)$. The example distills two general lessons.

First, the two responses divide the labor of identification. The forced-choice factors involve only utility differences, and they identify β from within-task attribute variation. The ordinal factors involve utility levels, locating each task’s $\bar{\mu}(\beta)$ against the cut points. This is the division that Proposition 2.1 formalizes. The forced choice identifies β , as in any multinomial logit, because the two goods in a task differ in x . The ordinal responses then identify the cut points, by locating each task’s $\bar{\mu}(\beta)$ on the utility scale.

Second, the meaning of a scale label is set-dependent. The label “unsure” is fixed for the consumer, since by construction it means $p \in [0.37, 0.87]$, but what the researcher should infer from it depends on the task: the expected purchase probability of a consumer who answers “unsure” is 0.80 in the attractive task and 0.60 in the unattractive one (Table 1, last row).⁵

2.6 Special Cases and Relation to Existing Models

Binary dual response, logistic link. Set $W = 2$ with the single cut point $c_1 = \gamma$. From Equation 9, $\Pr(\text{would buy}) = \Lambda(\bar{\mu}_{it} - \gamma) = S_{it} / (e^\gamma + S_{it})$, and the joint probabilities from Equation 11 reduce to:

$$\Pr(j, \text{buy}) = \frac{\exp(V_{itj})}{e^\gamma + S_{it}}, \quad \Pr(j, \text{not buy}) = \frac{\exp(V_{itj})}{S_{it}} \cdot \frac{e^\gamma}{e^\gamma + S_{it}}. \quad (13)$$

With $\gamma = 0$ these are exactly the joint probabilities of the “Unified Model” of Hung, Liu, et al. (2025) (their Eq. 4), which is in turn algebraically identical to the “DR-AnyMax” model of Diener et al. (2006); and Equation 13 with free γ is the standard single-stage MNL with a no-choice alternative-specific constant (Haaijer et al. 2001). The cumulative-link framework of Equation 10 thus contains the leading binary dual-response model exactly, as its two-point case

⁵The conditional expectation is $E[F(u^*) \mid u^* \in [c_1, c_2]]$ with $u^* \sim \text{Gumbel}(\bar{\mu}, 1)$, which has the closed form $\frac{S}{S+1} \left[e^{-(S+1)e^{-c_2}} - e^{-(S+1)e^{-c_1}} \right] / \left[e^{-Se^{-c_2}} - e^{-Se^{-c_1}} \right]$.

under the logistic link, and for $W > 2$ the additional cut points are the only new parameters. What the framework contains, however, our model does not reproduce: at $W = 2$ it predicts a different link, which is the content of the next paragraph and is testable on data already collected.

Binary dual response, our model. Set $W = 2$ in Equation 7: $\Pr(\text{not buy}) = \alpha_1^{S_{it}} = F(c_1 - \bar{\mu}_{it})$, a complementary log-log rather than logistic binary response. Our model therefore implies that even the familiar binary dual response is misspecified, if mildly given the proximity of the links, when consumers genuinely face outside-good uncertainty at the moment they answer. This is where the two accounts are furthest apart in practical terms and closest together numerically, and it converts a modelling disagreement into an exercise any reader can run: refit the second stage of an existing binary dual-response study with a complementary log-log link and compare.

DR-2Max. The other binary model in the literature, “DR-2Max” (Diener et al. 2006), as implemented in Sawtooth Software’s CBC/HB (Sawtooth Software 2017), models the second response as a binary logit on the *chosen good’s* deterministic utility: $\Pr(\text{buy} \mid j^*) = e^{V_{itj^*}} / (e^{V_{itj^*}} + e^\gamma)$. This requires drawing a fresh error for the chosen good in the second stage, so that the consumer who just declared j^* best re-evaluates it from scratch. Under our framework such conditioning is unnecessary: by Lemma 2.1(ii) the distribution of the best utility does not depend on which good attained it, so the second response should depend on the set only through $\bar{\mu}_{it}$. This distinction is testable, and the existing evidence favors the inclusive-value form, which our framework implies rather than assumes: Diener et al. (2006) found DR-AnyMax fit dual-response data better than DR-2Max, and Hung, Liu, et al. (2025) reach the same conclusion for their Unified Model.

Reductions to the binary model. An ordinal response can be reduced to a form the binary model accepts in two ways: (i) *dichotomization*, collapsing the scale at some point (e.g., top box or top two boxes) and estimating the binary model, which discards the ordinal information and hard-codes an arbitrary cut; or (ii) *constant-probability weighting*, mapping each scale point to a fixed purchase probability (e.g., “definitely would buy” = 0.9) independent of the choice set. Both are nested within, or dominated by, Equation 11: dichotomization is Equation 13 applied to a coarsened y_{it} , and constant-probability weighting violates Equation 7, under which the mapping from scale point to purchase probability necessarily depends on the attractiveness of the inside goods through S_{it} . These two reductions are the benchmarks in our simulation study (Section 3).

2.7 Identification

Because the second response depends on the choice set only through $\bar{\mu}_{it}$, and because the cut points sit on the same utility scale as $\bar{\mu}_{it}$, the model requires a location normalization, and given a suitable design it requires nothing further. This subsection makes the requirement precise. Fix the aggregate model ($\beta_i = \beta$) and let the data consist of tasks $t = 1, \dots, n$ with

inside-good attribute vectors $\{\mathbf{x}_{tj}\}_{j \in \mathcal{J}^+}$ and responses (j_t^*, y_t) generated by Equation 11 with parameter $\theta = (\beta, \mathbf{c}) \in \mathbb{R}^P \times \mathcal{C}$, where $\mathcal{C} = \{\mathbf{c} : c_1 < \dots < c_{W-1}\}$. Call θ and θ' *observationally equivalent* if they imply the same joint probabilities $\Pr_\theta(j_t^* = j, y_t = w)$ for every task design in the support of the design distribution, and let X denote the design matrix that stacks the rows $\mathbf{x}_{tj}'$ over all inside goods and tasks in that support.

Proposition 2.1. $\theta = (\beta, \mathbf{c})$ is identified if and only if the augmented matrix $[X \ \iota]$ has full column rank $P + 1$, where ι is a conformable vector of ones. Equivalently: the only $(v, \delta) \in \mathbb{R}^P \times \mathbb{R}$ satisfying $\mathbf{x}_{tj}'v = \delta$ for every inside good j and task t is $(v, \delta) = (\mathbf{0}, 0)$.

Proof. (Sufficiency.) Suppose θ and $\theta' = (\beta', \mathbf{c}')$ are observationally equivalent. Marginalizing the joint probabilities over y equates the MNL choice probabilities task by task, which implies $(\mathbf{x}_{tj} - \mathbf{x}_{tk})'(\beta - \beta') = 0$ for all j, k, t ; hence $\mathbf{x}_{tj}'(\beta - \beta') = a_t$, a constant within each task. Marginalizing over j equates the cumulative ordinal probabilities: $G(c_w - \bar{\mu}_t(\beta)) = G(c'_w - \bar{\mu}_t(\beta'))$ for all w, t . Since G is strictly increasing (for both links in Equation 10), $c_w - c'_w = \bar{\mu}_t(\beta) - \bar{\mu}_t(\beta')$ for all w, t ; the left side is free of t and the right side free of w , so both equal a constant δ . Direct computation gives $\bar{\mu}_t(\beta') = \ln \sum_j \exp(V_{tj} - a_t) = \bar{\mu}_t(\beta) - a_t$, so $a_t = \delta$ for every t ; that is, $\mathbf{x}_{tj}'(\beta - \beta') = \delta$ for all j, t . The rank condition then forces $\beta = \beta'$ and $\delta = 0$, hence $\mathbf{c} = \mathbf{c}'$. (*Necessity.*) If some $(v, \delta) \neq (\mathbf{0}, 0)$ satisfies $\mathbf{x}_{tj}'v = \delta$ everywhere, then $\theta' = (\beta + v, \mathbf{c} + \delta\iota)$ shifts every V_{tj} and every cut point by δ , leaving the MNL probabilities (a within-task constant drops out) and the ordinal probabilities ($G(c_w + \delta - (\bar{\mu}_t + \delta))$) unchanged, so $\theta' \neq \theta$ is observationally equivalent to θ . $\square$

The condition in miniature. Return to the two consumers of Section 2.5, with $\beta = 1$ and $\mathbf{c} = (0, 2)$, so that $\bar{\mu}_1 = 2.31$ and $\bar{\mu}_2 = -0.69$. Suppose the design had carried a constant: one extra attribute, equal to one for every good, with part-worth β_{const} . Setting $\beta_{\text{const}} = 1$ raises every V_{tj} by one and therefore raises both inclusive values by exactly one, to 3.31 and 0.31. Raise both cut points by one as well, to $\mathbf{c} = (1, 3)$, and every difference $c_w - \bar{\mu}_t$ returns to the value it had: $(-2.31, -0.31)$ for the first consumer and $(0.69, 2.69)$ for the second. The two ordinal distributions are therefore exactly what they were. The forced choice never saw the constant at all, since it turns on $\mathbf{x}_{t1} - \mathbf{x}_{t2}$. No quantity of data separates $(\beta_{\text{const}}, \mathbf{c}) = (0, (0, 2))$ from $(1, (1, 3))$, or from any other point along that ridge, and it is the cut points rather than the part-worths that do the absorbing.

Several remarks unpack the content of Proposition 2.1.

What the condition rules out. A violating direction (v, δ) is an attribute combination whose value is the same for every inside good ever shown. The leading case is an intercept: a column of ones in X makes $[X \ \iota]$ rank deficient by construction. The subtler case is a *hidden* intercept: if an attribute is dummy-coded with all L of its levels included, every alternative carries exactly one of them, so the dummies sum to ι and the same invariance appears (shift all L part-worths and all cut points equally). Attributes must therefore be coded against omitted reference levels (or effects-coded), and the design must contain no global constant.

This requirement is stricter than in the single-stage MNL with a no-choice alternative, where a full set of brand dummies is identified because the no-choice row of zeros breaks the constancy (Haaaijer et al. 2001). In the dual-response model the outside good never appears as a design row, since its role is absorbed by the cut points, so anything the design holds constant across inside goods is absorbed along with it.

What the normalizations do. The convention $\mathbf{x}_{t0} = \mathbf{0}$ with a Gumbel(0, 1) shock fixes the origin against which the cut points are measured: an outside-good alternative-specific constant would shift all cut points equally and is exactly the invariance in the proof, so it is subsumed rather than restricted. (Its visible trace is the binary case: at $W = 2$, the logistic link’s single cut point c_1 is the no-choice constant of Equation 13.) The unit Gumbel scale fixes the scale of β , as in any logit. Beyond these, the model imposes a genuine restriction rather than a normalization: the coefficient on $\bar{\mu}_{it}$ in Equation 10 is fixed at 1 by the theory, because both responses are driven by the same utility draw, so estimating a free slope λ on $\bar{\mu}_{it}$ and testing $\lambda = 1$ is an overidentification test of the behavioral model, not a required normalization.

Cut points versus the link. Given β (identified from the forced choice alone whenever X ’s within-task differences have full rank), the cut points are exactly identified even *without* variation in task attractiveness: $c_w = \bar{\mu} + G^{-1}(\Pr(y \leq w))$ pointwise. Variation in $\bar{\mu}_{it}$ across tasks therefore overidentifies $\mathbf{c}$, and it is what identifies the link: at a single value of $\bar{\mu}$, any two strictly increasing links fit the W category frequencies perfectly by relabeling cut points, so the shape of the link is identified only through how the ordinal distribution shifts as S_{it} varies. Designs that hold task attractiveness fixed cannot speak to it at all.

Population versus sample. Under Proposition 2.1 every scale category has strictly positive probability in every task (G is strictly increasing on $\mathbb{R}$ and the cut points are finite), so no support condition is needed for point identification. In finite samples, however, rarely used categories leave their adjacent cut points weakly identified, and near-violations of the rank condition (attribute combinations *nearly* constant across inside goods) manifest as an ill-conditioned information matrix. Both pathologies are visible in the smallest eigenvalues of the observed information, a diagnostic we report alongside the simulation results in Section 3.

The hierarchical model. With heterogeneous β_i , the argument applies respondent by respondent: individual-level identification requires the rank condition on each respondent’s own stacked design (their $T \times J$ rows), which rotating commercial designs typically satisfy. The hierarchy then supplies shrinkage, not identification of the model’s fixed features. In the heterogeneous-cut-point extension (Section 2.10.1) the same location tension reappears at the individual level, where a respondent-specific utility shift trades off against c_{i1} within that respondent’s likelihood, and is resolved by the same condition applied to the individual design, with the hierarchical prior doing real work when T is small.

Numerical verification. We verify Proposition 2.1 computationally (replication code accompanies the paper). Data simulated from the raw behavioral process (Gumbel utility draws, argmax, and the two reporting rules, not the derived likelihood) at three distinct parameter configurations for each model, on designs satisfying the rank condition, are recovered by maximum

likelihood: across all configurations the true parameter vector lies inside the 95% likelihood-ratio confidence region, the observed information is well-conditioned, repeated-sampling bias is statistically indistinguishable from zero, and model-based standard errors match the empirical sampling variation. Appending an intercept column, the canonical violation, produces exactly the predicted failure: $[X \ \iota]$ loses rank, the smallest eigenvalue of the observed information collapses to zero, and different starting values converge to different raw $(\beta_{\text{const}}, \mathbf{c})$ on the flat ridge while agreeing, to numerical precision, on the identified combinations $c_w - \beta_{\text{const}}$ and the non-constant coordinates of β . The same exercise confirms the factorization in Equation 11: the empirical joint distribution of (j^*, y) from one million behavioral draws is statistically indistinguishable from the product form.

2.8 Specification Tests

Because the theory restricts more than it parameterizes, the model is overidentified in two directions that convert directly into one-degree-of-freedom specification tests. Both are computed from the aggregate likelihood at trivial cost, and both have well-behaved operating characteristics (Section 3).

The unit-slope test. The behavioral model fixes the coefficient on the inclusive value in the second stage at exactly one, because the same utility draw generates both responses, so the natural one-parameter enlargement

$$\Pr(y_{it} \leq w) = G(c_w - \lambda \bar{\mu}_{it}) \quad (14)$$

nests the model at $\lambda = 1$. A free λ is identified by how the ordinal response distribution shifts as task attractiveness varies (Section 2.7), and departures are interpretable: $\lambda < 1$ indicates the second response under-reacts to the quality of the displayed set (consistent with inattention to the set or with scale-use compression), while $\lambda > 1$ indicates over-reaction. Testing $H_0: \lambda = 1$ by likelihood ratio or Wald is an overidentification test of the joint utility account, not a normalization check.

The conditioning test. The framework implies the second response depends on the choice set only through $\bar{\mu}_{it}$ (Lemma 2.1), whereas the DR-2Max tradition conditions on the chosen alternative's utility. The enlargement

$$\Pr(y_{it} \leq w) = G(c_w - [\bar{\mu}_{it} + \delta (V_{itj^*} - \bar{\mu}_{it})]) \quad (15)$$

spans the two: $\delta = 0$ is our model and $\delta = 1$ puts the full weight on the chosen option's deterministic utility, the DR-2Max predictor. Both points are testable against the estimated δ , so the long-standing 2Max-versus-AnyMax question (Diener et al. 2006; Hung, Liu, et al. 2025) becomes a parameter estimate rather than a model-selection exercise.

We apply both tests to the empirical data in Section 4. Their finite-sample size and power, including full power of the δ test against DR-2Max data at modest sample sizes, are documented in the simulation study.

2.9 The Information Content of the Ordinal Response

How much does a W -point second response add over a binary one? The question has a clean answer in terms of Fisher information, extending the efficiency analysis of Hung, Liu, et al. (2025) from binary to ordinal second responses.

The per-task Fisher information about β decomposes into the familiar MNL term (Huber and Zwerina 1996) plus a second-response increment. Writing $\pi_w = G(c_w - \bar{\mu}) - G(c_{w-1} - \bar{\mu})$ and $g = G'$, the second response contributes

$$\mathcal{I}_{it}^{(2)} = \omega(\bar{\mu}_{it}; \mathbf{c}) \bar{\mathbf{x}}_{it} \bar{\mathbf{x}}_{it}', \quad \omega = \sum_{w=1}^W \frac{[g(c_w - \bar{\mu}) - g(c_{w-1} - \bar{\mu})]^2}{\pi_w}, \quad (16)$$

where $\bar{\mathbf{x}}_{it} = \sum_{j \in \mathcal{J}^+} \Pr(j_{it}^* = j) \mathbf{x}_{tj}$ is the choice-probability-weighted average attribute vector (the gradient of $\bar{\mu}_{it}$ with respect to β). Two properties follow. First, the increment is positive semi-definite: the second response never destroys information, as in the binary result of Hung, Liu, et al. (2025). Second, ω is non-decreasing under refinement of the scale partition, since coarsening an ordinal response is a garbling in the sense of sufficiency. The binary response ($W = 2$) is therefore the least informative member of the family, and information rises toward the continuous-report limit as $W \rightarrow \infty$. Section 3 quantifies the practical magnitudes: how much of the binary-to-continuous gap a five- or seven-point scale captures, and how the gains vary with task attractiveness.

2.10 Heterogeneity

We accommodate preference heterogeneity with a standard hierarchical Bayesian specification (Rossi et al. 2005; Train 2001):

$$\beta_i \stackrel{iid}{\sim} \mathcal{N}(\bar{\beta}, \Sigma_\beta), \quad (17)$$

with conjugate priors on $(\bar{\beta}, \Sigma_\beta)$. In the main model the cut points $\mathbf{c}$ are common across consumers: the scale labels (“very likely,” “somewhat likely,” ...) are assumed to mean the same thing to everyone, while preferences vary.

2.10.1 Heterogeneous Cut Points

The assumption of a shared reporting scale is convenient but contestable: Greene and Hensher (2010) and Falco et al. (2015) document individual differences in the use of ordinal response scales. We therefore develop an extension with consumer-specific cut points. Parameterize

$$c_{i1} \in \mathbb{R}, \quad c_{iw} = c_{i,w-1} + \exp(\delta_{iw}) \text{ for } w = 2, \dots, W-1, \quad (18)$$

and let $\theta_i = (c_{i1}, \delta_{i2}, \dots, \delta_{i,W-1})'$ follow a multivariate normal population distribution, estimated jointly with (and, if desired, correlated with) β_i . The exponentiated increments enforce

monotonicity without constrained sampling. Identification at the individual level is necessarily partial, because with T tasks per consumer the $W - 1$ individual cut points and P individual preference parameters are jointly informed by only T ordinal reports. The hierarchy therefore matters here since it shrinks each respondent’s scale usage toward a population distribution that is estimated from the whole sample. Section 3 examines both the payoff to permitting cut-point heterogeneity when it is present in the data-generating process and the robustness of the common-cut-point model when it is not.

2.11 Estimation

The aggregate (homogeneous- β) model is estimated by maximum likelihood: the log of Equation 11 summed over consumers and tasks is smooth in $(\beta, \mathbf{c})$ after the monotonicity-preserving reparameterization of the cut points, and standard quasi-Newton methods converge quickly. Because the likelihood is a product of a choice component and an ordinal component, it can also be assembled in Apollo (Hess and Palma 2019) from that package’s standard building blocks, which we do as an independent check on our own code. Appendix E gives the details.

The hierarchical model is estimated with a custom Markov chain Monte Carlo sampler, implemented in R and following the hybrid Gibbs architecture of Rossi et al. (2005): (1) a random-walk Metropolis step for each β_i (and θ_i in the heterogeneous-cut-point extension) using the individual likelihood $\prod_t \ell_{it}$; (2) a random-walk Metropolis step for the common cut-point block on the unconstrained scale; and (3) conjugate draws for the population moments $(\bar{\beta}, \Sigma_{\beta})$. Model comparison uses the log marginal density approximated by the method of Newton and Raftery (1994), matching the practice in Hung, Liu, et al. (2025) and permitting direct comparisons among our model, the binary dual-response models, and the practitioner heuristics. Complete replication code accompanies the paper.

3 Simulation Study

Source: [Article Notebook](#)

The simulation program serves six purposes: verify that the estimators recover the model’s parameters and that the identification theory of Section 2.7 bites where it says it does; quantify what dichotomization costs, in the aggregate and, where it matters more, at the individual level; establish the operating characteristics of the specification tests of Section 2.8; probe robustness to the misspecifications a referee should worry about; turn the information results of Section 2.9 into design guidance; and attach economic stakes to the choice of estimator through a pricing counterfactual.

Two conventions apply throughout. All data are simulated from the *behavioral process*, meaning raw Gumbel utility draws, the argmax, and the reporting rules, never from the derived likelihood,

so parameter recovery validates the derivation and the code at the same time. And every experiment is seeded and reproducible.

Each subsection below summarizes one study and points to the notebook that contains it. The notebooks accompany the paper, run the experiments end to end, and are the source of the numbers reported here; they also carry the diagnostics, intermediate tables, and negative controls that are not worth article space. Readers who want to see a result rather than take it on trust should start there.

3.1 Design

The aggregate experiments use pooled tasks with $J = 4$ alternatives described by two 3-level attributes (dummy-coded against reference levels) and a continuous price-like attribute ($P = 5$), a $W = 5$ scale, across four parameter configurations that vary the location and spread of the response distribution. One is a *skewed* configuration with realistic bottom-heavy purchase intent (top-2-box share 18.4%), which is where the top-2-box convention is worst placed. The hierarchical experiments use $N = 300$ respondents, $T = 12$ estimation tasks plus 2 holdouts, $\beta_i \sim \mathcal{N}(\beta, \Sigma)$ with substantial heterogeneity, and the same attribute scheme.

3.2 Parameter Recovery and Identification

Aggregate. [Notebook 2](#) runs three parameter configurations per link at 20,000 tasks each. In every case the true parameter vector lies inside the 95% likelihood-ratio confidence region, the observed information is strongly positive definite, and the probability-scale cut points of our model recover to within a percentage point: a true $\alpha_1 = 0.02$ in the stress configuration is estimated at 0.0196. Repeated-sampling behavior matches the asymptotics. Across 40 independent datasets per link at $n = 5,000$, no parameter shows significant bias ($\max |t| = 2.05$ over 18 tests) and model-based standard errors match the empirical sampling variation (mean calibration ratio 1.008). Quadrupling n shrinks the largest observed bias in proportion, as $O(1/n)$ requires, which is what distinguishes ordinary finite-sample bias from a structural problem.

The same notebook carries two checks the analysis of Section 2.7 already anticipates, and which are therefore reported there rather than here. Appending an intercept, the canonical violation of Proposition 2.1, collapses the smallest eigenvalue of the observed information and produces the predicted flat ridge: different starting values reach identical likelihoods at different raw $(\beta_{\text{const}}, \mathbf{c})$ while agreeing to four decimal places on the identified combinations. And the empirical joint distribution of the two responses over one million behavioral draws is statistically indistinguishable from the product form of Equation 11, confirming the factorization that makes the likelihood tractable.

Hierarchical. [Notebook 3](#) establishes that the sampler of Section 2.11 recovers population means, covariances, and cut points on synthetic panels, and that its posterior uncertainty is

calibrated: across 8 replications, 90% central posterior intervals cover the truth at 89% and 83% respectively, with 50% intervals at 51% and 43%. Calibration matters independently of recovery, since a sampler can find the right answer and still misstate its own uncertainty. As an independent check of the hierarchy machinery, the forced-choice-only special case of our sampler agrees with an established hierarchical-MNL implementation on identical data to within 0.007 on every population parameter.

3.3 The Cost of Dichotomization

Collapsing the W -point response at a cut point yields a correctly specified binary dual-response model, so dichotomization costs efficiency and never consistency; [notebook 4](#) confirms this with bias tests in every cell. The comparison is therefore about precision, and it is computed two ways: exactly, from the Fisher information, and by paired Monte Carlo, fitting the ordinal model and the binary model to the *same* simulated datasets so that all differences are attributable to the coarsened second response.

Aggregate: modest at a lucky cut, material at an unlucky one. Table 2 reports exact variance ratios, equivalently the extra sample the binary analyst needs, for the top-2-box convention and for the worst fixed cut. Top-2-box dichotomization costs 4%–15% extra sample for the preference parameters while discarding 27%–52% of the second response’s own information; the worst cuts cost up to 33%. The analyst must commit to the cut before seeing the data, so the worst case is the relevant risk. The ordinal model is uniformly within about 2% of the full-information benchmark. Aggregate demand-*prediction* ratios are small (1.01–1.07) because prediction variance is dominated by the forced-choice stage that both estimators share, an honest caveat that the individual-level results overturn.

Table 2: Exact asymptotic variance ratios versus the full 5-point model (extra sample required for equal precision), and the share of the second response’s information each coarsening retains. Top-2-box=the literature’s default dichotomization. Full study in [notebook 4](#).

configuration	top-2-box, beta	worst cut, beta	omega share, top-2-box	omega share, worst
set1_moderate	1.041	1.091	0.73	0.38
set2_negative	1.088	1.119	0.59	0.40
set3_stress	1.152	1.327	0.59	0.19
set4_skewed	1.080	1.125	0.48	0.20

Source: [Article Notebook](#)

Individual level: the gaps widen. In the hierarchical setting, the model practitioners actually estimate, each respondent contributes only $T = 12$ ordinal reports and the pooled cut points no longer rescue the coarsened second stage. Across 20 paired replications, top-2-box

dichotomization inflates the variance of the shared cut point by a factor of 1.09, individual-level purchase-probability prediction error by 1.19 in variance terms, and individual part-worth RMSE by 4.2%, with identical forced-choice data. Holdout hit rates are statistically indistinguishable (0.511 versus 0.510), as they should be. The first response carries the ranking information, and what dichotomization damages is precisely the demand margin the dual response exists to measure.

3.4 Specification Tests: Size and Power

[Notebook 5](#) turns the two behavioral claims of [Section 2.8](#) into one-parameter extensions of the likelihood and establishes their operating characteristics over 300 (size) and 150 (power) replications per cell. Both tests hold their nominal 5% size and the estimators of λ and δ are unbiased ([Table 3](#)).

Two results deserve emphasis. The conditioning test rejects the DR-2Max second stage in *every* replication at both sample sizes, so the 2Max-versus-inclusive-value question does not require large data to settle. And the test is informative in both directions: when the data truly come from a DR-2Max process, $\hat{\delta}$ lands on 1 (mean 1.00) and our model is rejected with power 100%. Against slope alternatives the λ test has power 99% at 3,600 tasks but only 64% at 1,200, which is a design fact worth knowing before fielding a study intended to test the unit-slope claim.

Table 3: Rejection rates of nominal-5% likelihood-ratio specification tests. Under the null (top rows), the lambda and delta tests hold size while ‘delta=1’ shows power against DR-2Max; under the alternatives (bottom rows), the targeted test rejects while the others stay quiet. Full study in [notebook 5](#).

cell	n	reject lambda=1	reject delta=0	reject delta=1 (2Max)
size B	1200	0.067	0.050	1.000
size B	3600	0.063	0.027	1.000
size A	3600	0.047	0.033	1.000
power lambda=0.7	3600	0.993	0.053	1.000
power lambda=1.3	3600	0.993	0.060	1.000
power lambda=0.7	1200	0.640	0.053	1.000
power 2max (delta=1)	3600	0.067	1.000	0.073
power 2max (delta=1)	1200	0.060	1.000	0.053

Source: [Article Notebook](#)

3.5 Misspecification Robustness

[Notebook 6](#) breaks the model five ways, each time generating data from something the model does not describe and fitting the model anyway on hierarchical panels.

- **Wrong link.** Fitting the logistic link to data from our model and vice versa moves holdout purchase-probability predictions by at most 0.025 on the probability scale, versus 0.019 under the correct link, and WAIC selects the true link in 100% of replications at this design’s attractiveness spread. Link misspecification is a second-order concern for demand; it matters for extrapolation into the tails.
- **Ignored scale-usage heterogeneity.** With respondent-specific cut points in the data-generating process (dispersion 0.35 on c_{i1}), the common-cut model loses almost nothing: individual part-worth RMSE 0.532 versus 0.535 for the heterogeneous-cut model, with prediction and hit-rate differences equally negligible. At commercial task counts the extension of Section 2.10.1 is robustness insurance and a diagnostic, not a default, which is what the identification discussion anticipated.
- **Non-Gumbel errors.** With variance-matched normal first-stage errors, preference recovery and demand prediction are essentially unimpaired (individual RMSE 0.514, prediction RMSE 0.110); the misspecification is absorbed almost entirely by the cut-point locations, with the top-2-box cut shifting by 0.63.
- **Correlated alternatives.** Appendix D shows that the framework survives any GEV error structure once the inclusive value is redefined, so fitting the independent-Gumbel model to nested-logit data misspecifies exactly one thing: it uses $\ln \sum_j e^{V_j}$ where the truth is $\ln G(e^V)$. With four alternatives in two nests at dissimilarity 0.5 (within-nest error correlation 0.75), holdout demand prediction is unaffected (RMSE 0.110 against 0.111 under correct specification), and individual part-worth RMSE rises only to 0.553 from 0.524. The damage is again absorbed by the cut points, and in the direction theory predicts: nesting makes $G(e^V) < \sum_j e^{V_j}$, so the fitted model overstates the inclusive value and the top-2-box cut compensates upward by 0.21.
- **Task-order drift.** When the outside good drifts across the sequence (linear drift, and Hung-style probabilistic updating as in Hung, Kurz, et al. (2025)), the static model’s cut points absorb the average drift while part-worths remain accurate, and a simple order diagnostic, residual exceedance regressed on task position, detects the drift reliably (mean $|t| = 5.1$ under linear drift).

A pattern runs through all five: the damage lands on the cut points rather than on the preference parameters or the demand quantities built from them. The reason is structural rather than lucky. The forced choice identifies preferences and none of these misspecifications touch it, while the cut points are the model’s only free location and absorb whatever the second stage cannot express. The implication cuts both ways. Demand estimates are robust; the *structural* reading of the cut points as probability thresholds, which is what gives scale labels quantitative meaning, is the fragile part.

3.6 Designing the Scale

[Notebook 7](#) turns the information results of Section 2.9 into design guidance (Figure 1). With cut points at equal expected shares, a binary follow-up captures 46% of the continuous-report information limit and five points capture 78%; the increment beyond seven points is small. Cut *placement* matters as much as cut *count*: information-optimal cuts recover a further 15% over equal-share placement, and label sets that produce the skewed usage typical of fielded purchase-intent scales cost 29% relative to optimal.

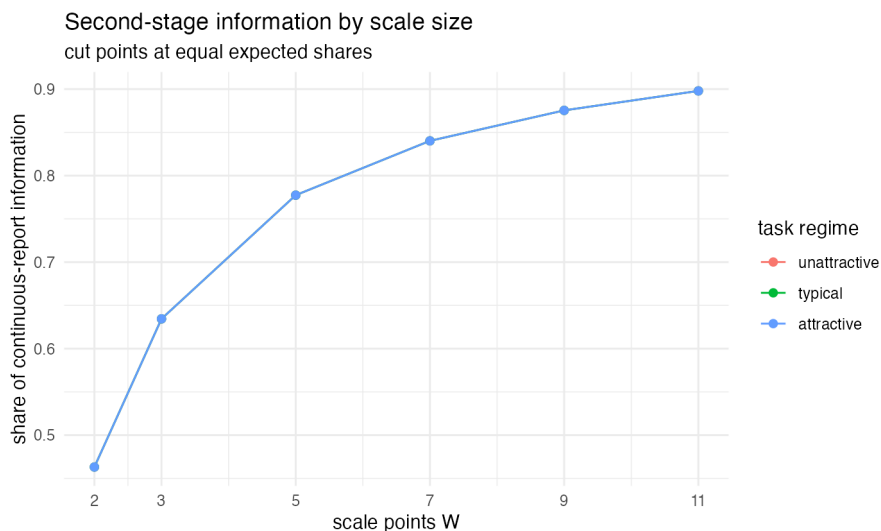


Figure 1: Share of the continuous-report information captured by a W -point second response (cut points at equal expected shares). The binary response is the least informative member of the family; most of the gap closes by five points. From notebook 7.

Source: [Article Notebook](#)

3.7 Economic Stakes: A Pricing Counterfactual

Finally, [notebook 8](#) pushes each estimator through the decision it exists to inform. A synthetic market of five brands with a quality attribute, wide price variation, and heterogeneous preferences calibrated to own-price elasticities of about -1.9 generates $W = 5$ dual-response data ($N = 500$, $T = 12$); each estimator's demand system then sets Nash equilibrium prices by damped best responses, verified from multiple starts. Demand is defined identically for all systems as top-2-box purchase, so the well-placed binary estimator is answering exactly the right question, and posterior-predictive demand uses full parameter draws rather than posterior means, which would flatten heterogeneity.

Table 4: Nash-pricing distortions by estimator: mean and max absolute equilibrium-price error versus the true equilibrium; error in the system’s own profit forecast at its equilibrium; and realized profit forgone relative to pricing at the true equilibrium. Full study in notebook 8.

system	mean price err %	max price err %	profit forecast err %	profit forgone %
ordinal	1.11	2.13	-2.98	0.37
bin_t2b	1.10	2.38	-2.47	-0.31
bin_top	2.57	5.73	-59.29	-0.84
heur_juster	4.84	8.52	105.82	2.18
heur_t2b	4.84	8.52	52.94	2.18

Source: [Article Notebook](#)

Three lessons (Table 4). First, the full ordinal model prices the market almost exactly. Second, the *well-placed* binary dual response also prices well, consistent with the efficiency analysis, but the analyst does not get to choose the placement after the fact: dichotomizing at the top box misprices by up to 5.7% and misstates its own profit forecast by 59%, because a top-box model can only answer a top-box question. Third, the constant-probability heuristic is the expensive shortcut. Its purchase incidence cannot respond to price, so it underprices systematically, its equilibrium being the inside-good-only equilibrium because the calibration constant cancels from every best response, and it overstates forecast profits by 106%. The heuristic’s failure is structural, not statistical: the constant is the problem, so no choice of constant and no amount of data repairs it.

4 Empirical Application

We illustrate the model with data from a commercial choice-based conjoint study conducted in partnership with a consumer insights consultancy, in which each choice task was followed by an ordinal purchase-likelihood question.

4.1 Data

4.2 Model Specifications and Estimation

We estimate four specifications on the same data: our model (Section 2) in aggregate and hierarchical form; the binary dual-response benchmark applied to the dichotomized scale (top-two-box); and the constant-probability heuristic. Hierarchical specifications use the sampler of Section 2.11.

Pre-specified test battery. Before any model comparison, we commit to the three specification tests of Section 2.8 on the aggregate fit: the unit-slope test ($\lambda = 1$), the conditioning test ($\delta = 0$), and the DR-2Max test ($\delta = 1$) — whose operating characteristics are established in Section 3.4 — together with the A-versus-B link comparison by holdout likelihood and WAIC. We also pre-commit to re-running the pricing counterfactual of Section 3.7 on the empirical posteriors, replacing the synthetic market with the study’s own category structure.

4.3 Results

4.4 Scale-Usage Heterogeneity

5 Discussion

5.1 What the Behavioral Model Changes

The premise that the consumer observes everything about the inside goods but not the outside-good shock does more than motivate a likelihood; it reorganizes how the dual-response literature’s models relate to one another. In the binary literature, the choice between DR-2Max and DR-AnyMax was framed as a question about the second question’s wording — whether the respondent evaluates her selected option or the whole set (Diener et al. 2006) — and the shared-error Unified Model of Hung, Liu, et al. (2025) was motivated by the internal consistency of a single utility draw. Our framework arrives at the inclusive-value form from the primitive assumption about information: because the maximum of Gumbel utilities is independent of which good attains it, *any* coherent report about the best option’s standing against the outside good must depend on the task only through its inclusive value. The empirical success of the AnyMax/Unified form over DR-2Max, documented in both Diener et al. (2006) and Hung, Liu, et al. (2025), is what this framework predicts. Where our account differs from Brazell et al. (2006) is in what the second response *is*: not a second choice from an augmented set, but a report about an unresolved comparison — which is what licenses asking for it on an ordinal scale in the first place.

The choice of link is, likewise, a substantive behavioral question dressed as a specification detail. It is also a question with a definite scope. For a binary follow-up both accounts are coherent and they disagree: resolving the outside-good draw first recovers the standard logit second stage, while our model implies a complementary log-log one even for yes/no data. Because the two diverge mainly in the tails, telling them apart requires either large samples or designs with wide variation in task attractiveness; Section 4 provides evidence from one commercial dataset, and we regard the question as open and worth testing on the many existing binary dual-response datasets, where our model’s binary case can be estimated with a one-line change to current code. For an ordinal follow-up the question does not arise in the same form, since a respondent who has resolved the outside good holds no graded likelihood to report.

5.2 Implications for Practice

Four points bear directly on applied work. First, if an ordinal dual response has been fielded, dichotomizing it is costly and arbitrary: the analyst’s choice of cut is made before seeing the data, the penalty ranges from a few percent of sample at a lucky cut to a third of it at an unlucky one (Section 3.3), the damage concentrates at the individual level where hierarchical estimation lives, and a dichotomized model can answer questions only at its own threshold — the pricing counterfactual’s top-box analyst misstates the profit forecast by more than half for exactly that reason. Second, constant-probability weighting has a *structural* flaw, not a statistical one: beyond the set-dependence of scale-point meaning (Equation 7 and Table 1), a fixed purchase incidence cancels out of every pricing first-order condition, so the heuristic’s price recommendations ignore the no-buy margin entirely — no amount of data fixes this. Third, the information results in Section 2.9 and Section 3.6 give concrete design guidance: five scale points capture most of the binary-to-continuous information gap (about 78%, versus 46% for binary), gains beyond seven points are small, and *where* the labels place the cut points matters as much as how many there are. Fourth, studies that intend to test the behavioral model (the link, the unit slope) should engineer variation in task attractiveness; designs without it cannot discriminate the links even in principle (Section 2.7), a fact our squashed-design experiment confirms empirically.

5.3 Relation to the Outside-Good Literature

Our model holds the outside good’s *mean* utility fixed at zero and locates all action in the consumer’s uncertainty about its realization. Two adjacent strands suggest natural extensions. Hung, Liu, et al. (2025) show that a budget constraint applied to the second response — but not the forced choice — improves fit and materially changes equilibrium price predictions; the same asymmetric treatment is available here, censoring the ordinal report for options priced above a respondent-level budget, and the ordinal response should sharpen the identification of that budget since near-threshold options will earn lukewarm rather than merely negative reports. Hung, Kurz, et al. (2025) document that the no-choice option’s utility can drift over the course of a survey as respondents learn about the category; in our notation this is drift in the cut points (or in the outside-good location), and the task-order diagnostics of Section 4.1 carry over directly. More broadly, the opt-out literature’s warning that a single constant rarely captures heterogeneous non-purchase behavior (Campbell and Erdem 2019; Haaijer et al. 2001) applies to the dual response as well; the heterogeneous-cut-point extension of Section 2.10.1 is one response, and richer structures (e.g., serial non-purchasers as a latent class) are compatible with the framework.

5.4 Limitations

Five limitations qualify the results. First, the Gumbel error assumption is load-bearing: the factorization of the joint likelihood and the inclusive-value form of the second response both rest on max-stability. Appendix D shows how far this can be pushed. Any generalized extreme value structure, nested logit included, preserves both properties once the inclusive value is redefined as the GEV generator, so correlation among the displayed alternatives costs the framework nothing. Error families outside that class do break the closed form: under a probit or a mixed logit with a correlated kernel the max utility has no closed-form distribution, and the second response would require simulation over it. Our robustness experiments soften the concern in one direction: with variance-matched normal errors, preference and demand estimates are essentially unimpaired, the misspecification being absorbed by the cut-point locations — so structural interpretation of the cut points deserves more caution than the demand quantities built from them. Second, we treat tasks as independent draws, abstracting from learning, fatigue, and reference formation across the sequence (Hung, Kurz, et al. 2025); the task-order diagnostic of Section 3.5 detects such drift reliably when present. Third, the common-cut-point model assumes a shared reading of the scale labels; the extension of Section 2.10.1 relaxes this, but individual-level identification is thin at commercial task counts, and our simulations find that ignoring moderate scale-usage heterogeneity costs little — the extension earns its keep as insurance and as a diagnostic, not as a default. Fourth, like all stated-preference methods, the model calibrates stated rather than revealed purchase behavior; the ordinal report inherits whatever hypothetical bias attends purchase-intent questions generally (Morrison 1979), and linking the estimated cut points to realized purchase rates — the ordinal analogue of the calibration literature on binary intentions — is a natural next step. Fifth, the likelihood presumes the two questions are asked together, as in the standard dual-response task. Designs that separate them, such as the separated dual response of Schlereth and Skiera (2017), break the coupling between the two responses and would require the stages to be modeled as separate draws. A separate methodological caution, documented in Appendix C, is that harmonic-mean marginal-likelihood estimates are optimistic; we therefore anchor model comparisons in holdout likelihood and WAIC alongside the conventional estimator.

6 Conclusion

Dual-response conjoint separates a question consumers answer readily, which of these do you like best, from one they cannot answer with certainty, would you actually buy it. This paper takes that uncertainty as the modeling primitive. A consumer who observes everything about the displayed alternatives but not her valuation of the outside good holds a purchase probability rather than a decision, so the natural follow-up question is ordinal and the natural model joins a multinomial logit for the forced choice to a cumulative-link ordinal model on the task’s inclusive value. The link function is not a modeling choice. It follows from whether the respondent reports the probability she still holds, as our model has her do, or resolves her uncertainty

and grades the comparison that results, and the two are distinguishable on data that vary the attractiveness of the displayed alternatives.

Three results follow. At two scale points the logistic link is the incumbent binary dual-response model, so the ordinal form is an extension of that literature rather than a departure from it, while our model's binary case is a complementary log-log response, which implies that the familiar yes/no follow-up is mildly misspecified and can be checked on data already collected. A finer scale never subtracts information, and five points capture most of the gap between a binary answer and a continuous one. The two ways an ordinal answer is otherwise reduced to something the binary model accepts, collapsing the scale at an arbitrary point and assigning each point a fixed purchase probability, are respectively wasteful and internally inconsistent, the second because the meaning of a scale label depends on what was on the screen. Simulation evidence confirms that the aggregate and hierarchical Bayesian estimators recover the model's parameters, including the cut points that give the scale labels quantitative meaning, and that the conclusions survive correlated alternatives, non-Gumbel errors, and drift in the outside good across a respondent's task sequence.

The broader point is that a response format and the model that reads it belong together. A finer scale is worth fielding only if something can use the extra categories, and a model is worth deriving only if the question it reads is one respondents can answer. Supplying the likelihood makes the ordinal follow-up a measurement instrument rather than a longer question, and it opens work on calibrating estimated cut points against realized purchase behavior, on using the ordinal response to identify budget constraints, and on the design of the scale itself.

Appendix A: Proofs

A.1 Proof of Lemma 2.1

Let $u_j = V_j + \eta_j$ with $\eta_j \stackrel{iid}{\sim} \text{Gumbel}(0, 1)$ for $j \in \mathcal{J}^+$, so u_j has CDF $F_j(t) = \exp(-e^{-(t-V_j)})$ and density $f_j(t) = e^{-(t-V_j)} \exp(-e^{-(t-V_j)})$. For any j and any u ,

$$\Pr(j^* = j, u^* \leq u) = \int_{-\infty}^u f_j(t) \prod_{k \neq j} F_k(t) dt = \int_{-\infty}^u e^{V_j} e^{-t} \exp(-e^{-t} S) dt,$$

using $\prod_{k \neq j} F_k(t) = \exp(-e^{-t} (S - e^{V_j}))$ with $S = \sum_k e^{V_k}$, and collecting terms. Substituting $s = e^{-t} S$ (so $ds = -e^{-t} S dt$) gives

$$\Pr(j^* = j, u^* \leq u) = \frac{e^{V_j}}{S} \int_{e^{-u} S}^{\infty} e^{-s} ds = \underbrace{\frac{e^{V_j}}{S}}_{\Pr(j^* = j)} \times \underbrace{\exp(-S e^{-u})}_{F_{\text{Gumbel}(\bar{\mu}, 1)}(u)}.$$

Letting $u \rightarrow \infty$ delivers the MNL choice probability; summing over j delivers $u^* \sim \text{Gumbel}(\bar{\mu}, 1)$ with $\bar{\mu} = \ln S$; and the factorization of the joint law for every (j, u) is precisely the independence of j^* and u^* . $\square$

The calculation above is short but gives little sense of why the two quantities separate. The following change of variable makes it transparent, and it is the same argument that generalizes to correlated alternatives in Appendix D.

Write $y_j = e^{V_j}$ and set $T_j = e^{-u_j}$. The map is decreasing, so it exchanges maxima for minima: the best good is the one with the smallest T_j , and $u^* = -\ln \min_j T_j$. Under the Gumbel assumption,

$$\Pr(T_j > t) = \Pr(u_j < -\ln t) = e^{-y_j t},$$

so each good becomes an independent waiting time that arrives at the constant rate y_j , and the consumer's chosen good is whichever arrives first.

For independent waiting times of this kind, the identity of the first arrival and the time of that arrival are independent, and the reason is that the rates do not change as time passes. At any instant, given that nothing has yet arrived, the chance that the next arrival is good j is y_j/S , the same at the start as at any later point. Which good arrives first is therefore settled entirely by the rates relative to one another, while how long the wait lasts is settled entirely by their sum S . Waiting longer says nothing about who will win. Transforming back, the shares y_j/S are the multinomial logit probabilities and the total S fixes the location of u^* , which is the content of Lemma 2.1. This is the division of labor between the two responses. The rates relative to one another are the utility differences that the forced choice reveals, and their sum is the inclusive value that the ordinal report reads, so the two responses read separate features of the same draw.

The same statement in utility terms is that the relative odds of being the maximizer are the same at every level of the maximum, because Gumbel upper tails decay exponentially and are therefore shift-invariant: $\Pr(u_1 > m)/\Pr(u_2 > m) = e^{V_1 - V_2}$ for every m .

This is a property of the extreme-value family and not of random utility models in general. Under normal errors with $V_1 > V_2$, the ratio of upper tails diverges as m grows, so a high realized maximum is disproportionately likely to have come from the better alternative. The level of the maximum then carries information about which alternative attained it, the joint law of (j^*, u^*) does not factor, and the counterpart of Equation 11 is unavailable. What Appendix D shows is that the factorization survives correlation among the inside goods, because the waiting times may be dependent and still arrive at rates whose ratios are constant over time; what fails outside the extreme-value family is the constancy of those ratios.

A.2 Derivation of the information increment (Equation 16)

The second response contributes the multinomial log-likelihood $\ell = \sum_w \mathbb{1}\{y = w\} \ln \pi_w$ with $\pi_w = G(c_w - \bar{\mu}) - G(c_{w-1} - \bar{\mu})$. Since $\partial \bar{\mu} / \partial \beta = \sum_j \Pr(j^* = j) \mathbf{x}_j = \bar{\mathbf{x}}$, the score with respect to

β in category w is $-[g_w - g_{w-1}]/\pi_w \cdot \bar{\mathbf{x}}$, writing $g_w = g(c_w - \bar{\mu})$ and $g = G'$ (with $g_0 = g_W = 0$). The information is the expected outer product of the score:

$$\mathcal{I}^{(2)} = \sum_{w=1}^W \pi_w \left(\frac{g_w - g_{w-1}}{\pi_w} \right)^2 \bar{\mathbf{x}} \bar{\mathbf{x}}' = \left[\sum_{w=1}^W \frac{(g_w - g_{w-1})^2}{\pi_w} \right] \bar{\mathbf{x}} \bar{\mathbf{x}}' = \omega(\bar{\mu}; \mathbf{c}) \bar{\mathbf{x}} \bar{\mathbf{x}}'. \quad \square$$

The joint information over $(\beta, \mathbf{c})$, including the cut-point blocks used by the estimation code, follows from the same score with $\partial\pi_w/\partial c_m = g_w \mathbb{1}\{m = w\} - g_{w-1} \mathbb{1}\{m = w-1\}$.

A.3 Monotonicity of ω under refinement

Let $\mathcal{C}'$ be a coarsening of the partition $\mathcal{C}$ obtained by merging adjacent categories w and $w+1$. The merged cell contributes $(g_{w+1} - g_{w-1})^2 / (\pi_w + \pi_{w+1})$ to ω' , while the two original cells contribute $(g_w - g_{w-1})^2 / \pi_w + (g_{w+1} - g_w)^2 / \pi_{w+1}$ to ω . Writing $a = g_w - g_{w-1}$, $b = g_{w+1} - g_w$, $u = \pi_w$, $v = \pi_{w+1}$, the Cauchy–Schwarz inequality in Engel form gives

$$\frac{(a+b)^2}{u+v} \leq \frac{a^2}{u} + \frac{b^2}{v},$$

so each merge weakly decreases ω , and any coarsening — being a finite sequence of adjacent merges — does the same. The binary response is the coarsest member of the family and hence the least informative; conversely, as the mesh of the partition goes to zero the interval score converges to the exact-observation score and $\omega \rightarrow \int (g')^2 / g dz$, the Fisher information of the latent’s location (equal to 1 for the standard Gumbel and 1/3 for the standard logistic). $\square$

Appendix B: Estimation Details

Aggregate model. The likelihood is maximized over the working parameters $(\beta, c_1, \ln(c_2 - c_1), \dots, \ln(c_{W-1} - c_{W-2}))$, which enforce cut-point monotonicity without constraints; standard errors for the cut points follow by the delta method. Interval probabilities are computed with tail-stable expressions $(\Lambda(h)\Lambda(-l)(1 - e^{l-h}))$ for the logistic link and $e^{-e^{-h}}(1 - e^{-(e^{-l} - e^{-h})})$ for the Gumbel link) so that extreme cut points do not lose precision to cancellation.

Hierarchical model. The sampler is a random-walk Metropolis-within-Gibbs in the style of Rossi et al. (2005) with three blocks per iteration: (1) all respondents’ β_i (and, in the heterogeneous-cut-point extension, θ_i) are updated simultaneously by independent normal random-walk proposals with covariance $s_i^2 \Sigma$; (2) the common cut-point block is updated by a random-walk proposal on the unconstrained scale; (3) the population moments $(\bar{\beta}, \Sigma)$ are drawn from their conjugate normal and inverse-Wishart conditionals. Priors: $\bar{\beta} \sim \mathcal{N}(\mathbf{0}, 100I)$, $\Sigma \sim \text{IW}(\text{dim}+3, (\text{dim}+3)I)$, and $\mathcal{N}(0, 100I)$ on the transformed cut points. Step sizes are

initialized at $2.93/\sqrt{\text{dim}}$ and adapted only during burn-in toward acceptance rates of 0.23 (individual blocks) and 0.30 (cut-point block). Unless noted, we run 30,000 iterations, discard 10,000, and keep every 10th draw.

The sampler was validated four ways (replication code accompanies the paper): parameter recovery on synthetic panels for both links; repeated-sampling coverage of 50% and 90% posterior intervals at nominal rates; agreement of the forced-choice-only special case with an independent hierarchical-MNL implementation (posterior means matching to less than 0.01 on all hyperparameters on common data); and convergence of the posterior toward the aggregate MLE when the data are generated without heterogeneity.

Appendix C: Marginal-Likelihood Estimation

Model comparisons in the paper report the log marginal density (LMD). For hierarchical models we follow the standard practice in this literature (Hung, Liu, et al. 2025) and use the harmonic-mean estimator of Newton and Raftery (1994), complemented by two quantities that do not rely on it: the WAIC computed from pointwise per-respondent likelihoods (Watanabe 2010; Vehtari et al. 2017) and the log posterior-predictive density of held-out tasks. The reason for the complements is that the harmonic-mean estimator is optimistic. In the aggregate model — where the parameter space is small enough for exact methods — we can quantify this directly: bridge sampling (Meng and Wong 1996; Gronau et al. 2020) and a Laplace approximation agree with each other to well under one log point, while the Newton–Raftery estimate exceeds them by dozens of log points, and trimming the smallest-likelihood draws widens rather than closes the gap:

Table 5: Aggregate-model log marginal density by estimator (simulated data, $n = 3,600$ tasks; from notebook 03).

estimator	log marginal density	vs. bridge
bridge	-9844.14	0.00
laplace	-9844.15	-0.01
NR	-9801.78	42.36
NR trim 5%	-9799.63	44.51
NR trim 20%	-9798.42	45.73

Source: [Article Notebook](#)

Because the bias is similar for models of similar dimension, Newton–Raftery *rankings* between the ordinal and binary dual-response models are typically preserved — but we treat the WAIC and holdout comparisons as primary whenever the two disagree.

Appendix D: Correlated Alternatives

Lemma 2.1 is stated for independent Gumbel errors, and the whole of Section 2 rests on it. This appendix shows that both of its properties hold for the entire generalized extreme value (GEV) family of McFadden (1981), so that correlation among the inside goods costs the framework nothing beyond a redefinition of the inclusive value.

Let the inside-good components have the GEV joint distribution

$$\Pr(\eta_1 \leq e_1, \dots, \eta_J \leq e_J) = \exp(-G(e^{-e_1}, \dots, e^{-e_J})), \quad (19)$$

where the generator $G : \mathbb{R}_+^J \rightarrow \mathbb{R}_+$ is homogeneous of degree one, tends to infinity in each argument, and has mixed partial derivatives that alternate in sign. Independent Gumbel errors are the special case $G(y) = \sum_j y_j$. Write $y_j = \exp(V_j)$, $G_j = \partial G / \partial y_j$, and

$$S^G := G(y_1, \dots, y_J), \quad \bar{\mu} := \ln S^G. \quad (20)$$

Lemma 6.1. *Under Equation 19, with $u_j = V_j + \eta_j$ and $u^* = \max_j u_j$:*

- (i) $u^* \sim \text{Gumbel}(\bar{\mu}, 1)$ with $\bar{\mu} = \ln S^G$;
- (ii) j^* and u^* are statistically independent, with $\Pr(j^* = j) = y_j G_j(y) / S^G$.

Proof. The joint CDF of the utilities is

$$H(m_1, \dots, m_J) = \exp(-G(y_1 e^{-m_1}, \dots, y_J e^{-m_J})).$$

For (i), set every $m_j = m$ and use homogeneity of degree one: $G(y e^{-m}) = e^{-m} G(y)$, so $H(m, \dots, m) = \exp(-e^{-m} S^G)$, which is the Gumbel $(\ln S^G, 1)$ CDF.

For (ii), write $t_k = y_k e^{-m_k}$ and differentiate:

$$\frac{\partial H}{\partial m_j} = \exp(-G(t)) G_j(t) y_j e^{-m_j}.$$

Because G is homogeneous of degree one, G_j is homogeneous of degree zero, so evaluating at $m_k = s$ for every k gives $G_j(t) = G_j(y)$ and $G(t) = e^{-s} S^G$. Hence

$$\left. \frac{\partial H}{\partial m_j} \right|_{m_k = s \ \forall k} = \exp(-e^{-s} S^G) G_j(y) y_j e^{-s} = \frac{y_j G_j(y)}{S^G} \cdot \frac{d}{ds} \exp(-e^{-s} S^G),$$

and integrating from $-\infty$ to u ,

$$\Pr(j^* = j, u^* \leq u) = \underbrace{\frac{y_j G_j(y)}{S^G}}_{\Pr(j^* = j)} \times \underbrace{\exp(-e^{-u} S^G)}_{\Pr(u^* \leq u)}$$

for every j and every u , which is the claimed independence. Letting $u \rightarrow \infty$ recovers the GEV choice probability. $\square$

Everything in Section 2 follows with S_{it} replaced by S_{it}^G . The first response becomes the GEV choice probability rather than the multinomial logit; the second response is unchanged in form, since Equation 7, Equation 9, and the cumulative-link representation Equation 10 depend on the choice set only through $\bar{\mu}_{it}$; and the factorization Equation 11 holds for the same reason as before. The location invariance behind Proposition 2.1 also survives: shifting every V_{itj} by δ multiplies S_{it}^G by e^δ and so shifts $\bar{\mu}_{it}$ by exactly δ , which the cut points absorb. The rank condition must then be read against the design as it enters G rather than a simple sum, and the nesting result of Section 2.6 becomes $\Pr(\text{buy}) = S_{it}^G / (e^\gamma + S_{it}^G)$.

The leading application is nested logit, $G(y) = \sum_n \left(\sum_{j \in n} y_j^{1/\lambda_n} \right)^{\lambda_n}$, in which alternatives within a nest are correlated whenever $\lambda_n < 1$. We confirm Lemma 6.1 numerically for a five-alternative, two-nest design with $\lambda = (0.5, 0.8)$ using the positive-stable mixture representation of the nested logit (`R/homogeneous/gev_check.R`). Across two million draws the simulated distribution of u^* matches Gumbel($\ln S^G, 1$) (location 1.845 against 1.846 analytically; Kolmogorov-Smirnov $p = 0.17$), the simulated choice shares match $y_j G_j / S^G$ to four decimal places, and the conditional distribution of u^* given j^* is indistinguishable from its unconditional counterpart at every quartile (chi-square test of independence, $p = 0.70$).

Two limits of the result are worth stating. It covers correlation among the *inside* goods only: the outside-good shock η_{it0} is assumed independent of them throughout, and The logistic link uses that independence to obtain $d_{it} \sim \text{Logistic}(\bar{\mu}_{it}, 1)$. And it does not extend to error structures outside the GEV class, such as multinomial probit or a mixed logit with correlated kernel errors, for which u^* has no closed-form distribution and the factorization that makes Equation 11 tractable is unavailable.

Appendix E: Implementation in Apollo

Source: [Article Notebook](#)

The replication archive implements the model twice: once directly, in the routines described in Appendix B, and once in Apollo (Hess and Palma 2019), the freeware choice-modelling package for R. The second implementation serves two purposes. It cross-validates our code against an independently developed and widely used toolkit, and it shows how the model fits into an estimation framework readers are likely to already have installed.

The model as a two-component structure. Apollo builds a likelihood by multiplying components. Our joint likelihood Equation 11 is exactly of that form, so the forced choice is an `apollo_mnl` component and the ordinal follow-up is a second component over the same tasks. This is the architecture of the hybrid choice models for which the package is often used, in which a multinomial logit for the choice is combined with ordered-response components for attitudinal indicators (Daly et al. 2012). The resemblance is worth being precise about, because the difference is the content of Section 2. In a hybrid choice model the ordinal component is

driven by a latent attitude that is exogenous to the choice, and it is connected to the choice utilities through freely estimated loading parameters. Here the ordinal component is driven by the inclusive value of the same choice set, and its loading is fixed at one rather than estimated. Our model is the restricted case, and Lemma 2.1 is what licenses the restriction.

The logistic specification is native; ours is not. With the second stage a cumulative logit in $\bar{\mu}_{it}$, The logistic specification is assembled from stock parts: `apollo_mnl` for the choice, `apollo_ol` for the follow-up with the task’s inclusive value passed as its single explanatory variable and the cut points as its thresholds, and `apollo_combineModels` to join them. Our model has no stock part. Apollo supplies ordered components for the logistic and normal links and no others, so the complementary-log-log form Equation 7 must be written out and passed through `apollo_ownModel`, the hook for user-defined likelihoods. This is a small piece of code, and once supplied it inherits the rest of the package. The asymmetry is worth noting for its own sake: the standard toolkit anticipates the behavior in which the respondent resolves her uncertainty before answering, and not the behavior in which she reports the probability she still holds.

Agreement. Table 6 reports the largest discrepancy between the two implementations over 5 independent datasets of 2,000 tasks in each of two configurations, with both implementations started from the same values.

Table 6: Largest absolute discrepancy between our maximum-likelihood routine and the same model estimated in Apollo, over five independent datasets per configuration (from `R/apollo/compare_apollo.R`).

configuration	ordinal component	beta	cut points	standard errors	log- likelihood
Our model, $W = 5$	<code>apollo_ownModel</code>	1e-06	2e-06	1e-07	9e-10
Logistic, $W = 2$ (binary)	<code>apollo_ol</code>	5e-07	5e-07	5e-09	6e-11

Source: [Article Notebook](#)

Point estimates agree to 2e-06 or better, standard errors to 1e-06, and the maximized log-likelihood to 9e-10, which is convergence tolerance rather than disagreement. The two implementations share no code: Apollo computes the ordinal probabilities for our model from the direct expression $e^{-e^{-(c_w - \bar{\mu})}} - e^{-e^{-(c_{w-1} - \bar{\mu})}}$, while ours uses the tail-stable rearrangement of Appendix B, and the cut points enter Apollo’s likelihood as free parameters rather than through the monotone increment parameterization. The binary row is the incumbent dual-response model of Diener et al. (2006) and Hung, Liu, et al. (2025), recovered as the two-point special case of Section 2.6.

Why the hierarchical model is not estimated in Apollo. Apollo reaches Bayesian estimation by wrapping `RSGB`, which requires each random parameter to be assigned one

of a fixed menu of marginal distributions: normal, log-normal, censored normal, or Johnson SB. None of them can express an ordered vector. The heterogeneous-cut-point extension of Section 2.10.1 gives each respondent her own thresholds θ_i , which must stay ordered for the likelihood to be defined, and we impose that through a transformation to an unconstrained scale inside a random-walk block. That is expressible in a purpose-written sampler and not in the RSGHB menu. The restriction is specific to the Bayesian path: `apollo_01` accepts thresholds that vary across observations, so classical mixed-threshold specifications are available. This is the reason the hierarchical results in Section 3 come from the sampler of Appendix B rather than from Apollo.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic), accessed through Claude Code, iteratively in the drafting and editing of manuscript text and select coding tasks. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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